

Reflection on Web and Mobile Application Development

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Gains from Development

1. Technical Improvement

- **Frontend Framework Usage:**
By using React Native to develop the Blog app, I gained a deeper understanding of frontend development. React Native offers rich components and an efficient development model, allowing for quick interface setup. For example, the Stack.Screen component was used for routing and setting page headers, improving navigation logic clarity.
- **State Management:**
I learned to use React Hooks like useState and useEffect to manage state and side effects in components. This was crucial for handling user input, data loading, and page updates. On the login page, useState managed email and password inputs and controlled login state.
- **Async Operations:**
Many app features required async tasks like API calls and file uploads. By using async/await, I could handle these tasks elegantly and avoid callback hell, such as when fetching blog details and user info.

2. Project Management & Architecture Design

- **Modular Development:**
The app was divided into components like blog list, blog details, and user profile, which improved code clarity and reusability.
- **API Design and Usage:**
I learned to design and call APIs and understood frontend-backend interaction. With the ky library, I could easily send requests to the server. I also implemented error handling and retry logic for better reliability.

3. Problem-Solving Skills

During development, I faced various issues and learned to resolve them independently by reading docs, searching online, and asking for help. For instance, I had trouble uploading images due to file format and path issues, which I resolved through debugging and research.

Problems Encountered

1. Compatibility Issues

- **Device Adaptability:**
Different devices caused layout/display inconsistencies. I addressed this by using relative units and responsive layouts.
- **OS Compatibility:**
Some Expo features behaved differently on older Android/iOS versions. Testing and adjustments were needed per version.

2. Performance Issues

- **Network Request Bottlenecks:**
Too many requests or large data loads slowed down responses. I optimized requests with caching and pagination.
- **Memory Management:**
Loading large datasets in ScrollView could cause memory leaks. Monitoring and releasing unused resources was necessary.

3. Feature Implementation Challenges

- **Complex Interactions:**
Features like swipe-to-delete, likes, and favorites required deeper understanding of gestures and state management.
- **Permission Handling:**
Features involving camera/gallery required proper permission management and a balance of UX and security.

Conclusion

Developing the Blog app helped me grow in technical skills, project organization, and problem-solving. Despite challenges, I gained deeper insights into programming principles. I will continue to improve the app and enhance user experience while learning new tools and technologies.